Part III

Processing by application of heat

Heat treatment remains one of the most important methods used in food processing, not only because of the desirable effects on eating quality (many foods are consumed in a cooked form and processes such as baking produce flavours that cannot be created by other means), but also because of the preservative effect on foods by the destruction of enzymes, micro-organisms, insects and parasites. The other main advantages of heat processing are:

1. relatively simple control of processing conditions
2. capability to produce shelf-stable foods that do not require refrigeration
3. destruction of anti-nutritional factors (e.g. trypsin inhibitor in some legumes)
4. improvement in the availability of some nutrients (e.g. improved digestibility of proteins, gelatinisation of starches and release of bound niacin).

However, heat also alters or destroys components of foods that are responsible for their individual flavour, colour, taste or texture and as a result they are perceived to have a lower quality and lower value. Fortunately the differences in D-values (Chapter 1) between these components and micro-organisms or enzymes can be exploited using higher temperatures and shorter times in heat processing. High-temperature short-time (HTST) processing can be designed to produce the same level of microbial or enzyme destruction at lower temperatures for longer times, but the sensory characteristics and nutritional value of foods is substantially retained. Developments in blanching (Chapter 10), pasteurisation (Chapter 11), heat sterilisation (Chapter 12), evaporation (Chapter 13) and dehydration (Chapter 15) have each focused on improved technology and better control of processing conditions to achieve higher quality products. Extrusion (Chapter 14) is by its nature a HTST process and other processes, including dielectric and ohmic heating (Chapter 18) are designed to cause minimal damage to the quality of foods.

Other more severe heat processes, including baking, roasting (Chapter 16) and frying (Chapter 17) are intended to change the sensory characteristics of a product, and preservation is achieved by either further processing (e.g. chilling or freezing, Chapters 19 and 21) or by selection of suitable packaging systems (Chapters 20, 24 and 25).

Another important effect of heating is the selective removal of volatile components from a food. In evaporation (Chapter 13) and dehydration (Chapter 15), the removal of water inhibits microbial growth and enzyme activity and thus achieves preservation. In distillation (Chapter 13) either alcohol is selectively removed to produce concentrated spirits, or flavour components are recovered and added back to foods to improve their sensory characteristics.

A. Heat processing using steam or water

10

Blanching

Blanching serves a variety of functions, one of the main ones being to destroy enzymic activity in vegetables and some fruits, prior to further processing. As such, it is not intended as a sole method of preservation but as a pre-treatment which is normally carried out between the preparation of the raw material (Chapter 3) and later operations (particularly heat sterilisation, dehydration and freezing (Chapters 12, 15 and 21)). Blanching is also combined with peeling and/or cleaning of food (Chapter 3), to achieve savings in energy consumption, space and equipment costs.

A few processed vegetables, for example onions and green peppers, do not require blanching to prevent enzyme activity during storage, but the majority suffer considerable loss in quality if blanching is omitted or if they are under-blanched. To achieve adequate enzyme inactivation, food is heated rapidly to a pre-set temperature, held for a pre-set time and then cooled rapidly to near ambient temperatures. The factors which influence blanching time are:

- type of fruit or vegetable
- size of the pieces of food
- blanching temperature
- method of heating.

10.1 Theory

The theory of unsteady-state heat transfer by conduction and convection, which is used to calculate blanching time, and a sample problem (Sample problem 1.7) are described in Chapter 1.

The maximum processing temperature in freezing and dehydration is insufficient to inactivate enzymes. If the food is not blanched, undesirable changes in sensory characteristics and nutritional properties take place during storage. In canning, the time taken to reach sterilising temperatures, particularly in large cans, may be sufficient to allow enzyme activity to take place. It is therefore necessary to blanch foods prior to

these preservation operations. Under-blanching may cause more damage to food than the absence of blanching does, because heat, which is sufficient to disrupt tissues and release enzymes, but not inactivate them, causes accelerated damage by mixing the enzymes and substrates. In addition, only some enzymes may be destroyed which causes increased activity of others and accelerated deterioration.

The heat resistance of enzymes is characterised by D and z values (Chapter 1). Enzymes which cause a loss of eating and nutritional qualities in vegetables and fruits include lipoxygenase, polyphenoloxidase, polygalacturonase and chlorophyllase. Two heat-resistant enzymes which are found in most vegetables are catalase and peroxidase. Although they do not cause deterioration during storage, they are used as marker enzymes to determine the success of blanching. Peroxidase is the more heat resistant of the two, so the absence of residual peroxidase activity would indicate that other less heat-resistant enzymes are also destroyed. The factors that control the rate of heating at the centre of the product are discussed in Chapter 1 and can be summarised as:

- the temperature of the heating medium
- the convective heat transfer coefficient
- the size and shape of the pieces of food
- the thermal conductivity of the food.

Blanching reduces the numbers of contaminating micro-organisms on the surface of foods and hence assists in subsequent preservation operations. This is particularly important in heat sterilisation (Chapter 12), as the time and temperature of processing are designed to achieve a specified reduction in cell numbers. If blanching is inadequate, a larger number of micro-organisms are present initially and this may result in a larger number of spoiled containers after processing. Freezing and drying do not substantially reduce the number of micro-organisms in unblanched foods and these are able to grow on thawing or rehydration.

Blanching also softens vegetable tissues to facilitate filling into containers and removes air from intercellular spaces which increases the density of food and assists in the formation of a head-space vacuum in cans (Chapters 12 and 25).

10.2 Equipment

The two most widespread commercial methods of blanching involve passing food through an atmosphere of saturated steam or a bath of hot water. Both types of equipment are relatively simple and inexpensive. Microwave blanching is not yet used commercially on a large scale. It is discussed further in Chapter 18. There have been substantial developments to blanchers in recent years to reduce the energy consumption and also to reduce the loss of soluble components of foods, which reduces the volume and polluting potential of effluents (Chapter 26) and increases the yield of product.

The yield[1] of food from the blanching operation is the most important factor in determining the commercial success of a particular method. In some methods the cooling stage may result in greater losses of product or nutrients than the blanching stage, and it is therefore important to consider both blanching and cooling when comparing different methods. Steam blanching results in higher nutrient retention provided that cooling is by cold-air or cold-water sprays. Cooling with running water (fluming) substantially

1. Weight of food after processing compared to the weight before processing.

increases leaching losses,[2] but the product may gain weight by absorbing water and the overall yield is therefore increased. Air cooling causes weight loss of the product due to evaporation, and this may outweigh any advantages gained by nutrient retention (Bomben *et al.*, 1975).

There are also substantial differences in yield and nutrient retention due to differences in the type of food and differences in the method of preparation (for example slicing and peeling (Chapter 3) increase losses and reduce the yield).

Recycling of water does not affect the product quality or yield but substantially reduces the volume of effluent produced. However, it is necessary to ensure adequate hygienic standards for both the product and equipment by preventing a build-up of bacteria in cooling water, and the improved hygiene control may result in additional costs which outweigh savings in energy and higher product yield.

10.2.1 Steam blanchers

The advantages and limitations of steam blanchers are described in Table 10.1. In general this is the preferred method for foods with a large area of cut surfaces as leaching losses are much smaller than those found using hot-water blanchers.

At its simplest a steam blancher consists of a mesh conveyor belt that carries food through a steam atmosphere in a tunnel. The residence time of the food is controlled by the speed of the conveyor and the length of the tunnel. Typically a tunnel is 15 m long and 1–1.5 m wide. The efficiency of energy consumption is 19% when water sprays are used at the inlet and outlet to condense escaping steam. Alternatively, food may enter and leave the blancher through rotary valves or hydrostatic seals to reduce steam losses and increase energy efficiency to 27%, or steam may be re-used by passing through Venturi valves. Energy efficiency is improved to 31% using combined hydrostatic and Venturi devices (Scott *et al.*, 1981).

In conventional steam blanching, there is often poor uniformity of heating in the multiple layers of food. The time–temperature combination required to ensure enzyme inactivation at the centre of the bed results in overheating of food at the edges and a consequent loss of texture and other sensory characteristics. Individual quick blanching

Table 10.1 Advantages and limitations of conventional steam and hot-water blanchers

Equipment	Advantages	Limitations
Conventional steam blanchers	Smaller loss of water-soluble components. Smaller volumes of waste and lower disposal charges than water blanchers, particularly with air cooling instead of water. Easy to clean and sterilise	Limited cleaning of the food so washers also required. Uneven blanching if the food is piled too high on the conveyor. Some loss of mass in the food.
Conventional hot-water blancher	Lower capital cost and better energy efficiency than steam blanchers	Higher costs in purchase of water and charges for treatment of large volumes of dilute effluent (Chapter 26). Risk of contamination by thermophilic bacteria.

2. Washing of soluble components from the food.

(IQB) which involves blanching in two stages, was developed to overcome this problem (Lazar *et al.*, 1971). In the first stage the food is heated in a single layer to a sufficiently high temperature to inactivate enzymes. In the second stage (termed *adiabatic holding*) a deep bed of food is held for sufficient time to allow the temperature at the centre of each piece to increase to that needed for enzyme inactivation. The reduced heating time (for example 25 s for heating and 50 s for holding 1 cm diced carrot compared with 3 min for conventional blanching), results in an improvement in the efficiency of energy consumption to 86–91% (Cumming *et al.*, 1984). The mass of product blanched per kilogram of steam increases from 0.5 kg per kilogram of steam in conventional steam blanchers to 6–7 kg per kilogram of steam, when small-particulate foods (for example peas, sliced or diced carrots) are blanched.

Nutrient losses during steam blanching are reduced by exposing the food to warm air (65ºC) in a short preliminary drying operation (termed '*pre-conditioning*'). Surface moisture evaporates and the surfaces then absorb condensing steam during IQB. Weight losses are reduced to 5% of those found using conventional steam blanching (Lazar *et al.*, 1971). Pre-conditioning and individual quick blanching are reported to reduce nutrient losses by 81% for green beans, by 75% for Brussels sprouts, by 61% for peas and by 53% for lima beans and there is no reduction in the yield of blanched food (Bomben *et al.*, 1973).

The equipment for IQB steam blanching (Fig. 10.1(a)) consists of a bucket elevator which carries the food to a heating section. The elevator is located in a close fitting tunnel to reduce steam losses. A single layer of food is heated on a conveyor belt and then held on a holding elevator before cooling. The cooling section employs a fog spray to saturate the cold air with moisture. This reduces evaporative losses from the food and reduces the amount of effluent produced. Typically the equipment processes up to 4500 kg h^{-1} of food. The complete inactivation of peroxidase is achieved with a minimum loss in quality, indicated by the retention of 76–85% of ascorbic acid.

Batch *fluidised-bed blanchers* operate using a mixture of air and steam, moving at approximately 4.5 m s^{-1}, which fluidises and heats the product simultaneously. The design of the blanching chamber promotes continuous and uniform circulation of the food until it is adequately blanched. Although these blanchers have not yet been widely used at a commercial scale, they are reported to overcome many of the problems associated with both steam and hot-water methods (Gilbert *et al.*, 1980). The advantages include:

- faster, more uniform heating
- good mixing of the product
- a substantial reduction in the volume of effluent
- shorter processing times and hence smaller losses of vitamins and other soluble heat sensitive components of food.

A continuous fluidised-bed blancher is described by Philippon (1984).

10.2.2 Hot-water blanchers

There are a number of different designs of blancher, each of which holds the food in hot water at 70–100ºC for a specified time and then removes it to a dewatering-cooling section. The advantages and limitations of hot-water blanchers are described in Table 10.1.

In the widely used *reel blancher*, food enters a slowly rotating cylindrical mesh drum which is partly submerged in hot water. The food is moved through the drum by internal flights. The speed of rotation and length control the heating time. *Pipe blanchers* consist

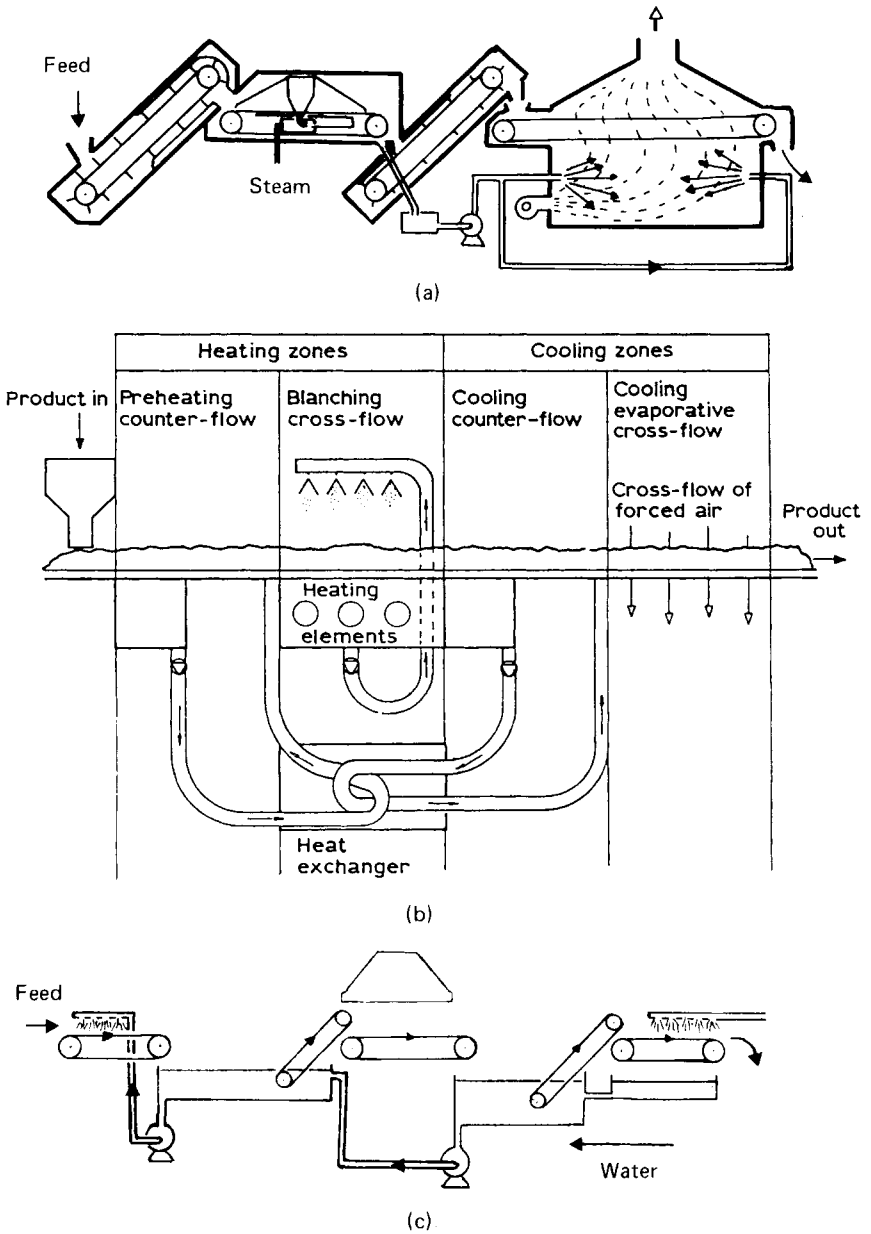


Fig. 10.1 Blanchers: (a) IQB steam blancher (after Timbers *et al.* (1984)); (b) blancher–cooler (from Hallstrom *et al.* (1988)) and (c) counter-current blancher (after Wendt *et al.* (1983)).

of a continuous insulated metal pipe fitted with feed and discharge ports. Hot water is recirculated through the pipe and food is metered in. The residence time of food in the blancher is determined by the length of the pipe and the velocity of the water. These blanchers have the advantage of a large capacity while occupying a small floor space. In some applications they may be used to transport food simultaneously through a factory.

Developments in hot-water blanchers, based on the IQB principle, reduce energy consumption and minimise the production of effluent. For example, the *blancher-cooler*

has three sections: a pre-heating stage, a blanching stage and a cooling stage (Fig. 10.1(b)). The food remains on a single conveyor belt throughout each stage and therefore does not suffer the physical damage associated with the turbulence of conventional hot-water blanchers. The food is pre-heated with water that is recirculated through a heat exchanger. After blanching, a second recirculation system cools the food. The two systems pass water through the same heat exchanger, and this heats the pre-heat water and simultaneously cools the cooling water. Up to 70% of the heat is recovered. A recirculated water-steam mixture is used to blanch the food, and final cooling is by cold air. Effluent production is negligible and water consumption is reduced to approximately $1\,m^3$ per 10 t of product. The mass of product blanched is 16.7–20 kg per kilogram of steam, compared with 0.25–0.5 kg per kilogram in conventional hot-water blanchers. An alternative design, used for blanching broccoli, lima beans, spinach and peas, is described by Wendt *et al.* (1983) in which water and food move counter-currently (Fig. 10.1(c)).

10.3 Effect on foods

The heat received by a food during blanching inevitably causes some changes to sensory and nutritional qualities. However, the heat treatment is less severe than for example in heat sterilisation (Chapter 12), and the resulting changes in food quality are less pronounced. In general, the time–temperature combination used for blanching is a compromise which ensures adequate enzyme inactivation but prevents excessive softening and loss of flavour in the food (Fig. 10.2).

10.3.1 Nutrients

Some minerals, water-soluble vitamins and other water-soluble components are lost during blanching. Losses of vitamins are mostly due to leaching, thermal destruction and, to a lesser extent, oxidation. The extent of vitamin loss depends on a number of factors including:

- the maturity of the food and variety
- methods used in preparation of the food, particularly the extent of cutting, slicing or dicing

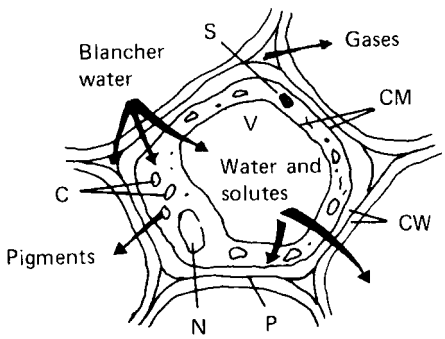


Fig. 10.2 Effect of blanching on cell tissues: S, starch gelatinised; CM, cytoplasmic membranes altered; CW, cell walls little altered; P, pectins modified; N, nucleus and cytoplasmic proteins denatured; C, chloroplasts and chromoplasts distorted.

Table 10.2 Effect of blanching method on ascorbic acid losses in selected vegetables

Treatment	Loss (%) of ascorbic acid		
	Peas	Broccoli	Green beans
Water blanch–water cool	29.1	38.7	15.1
Water blanch–air cool	25.0	30.6	19.5
Steam blanch–water cool	24.2	22.2	17.7
Steam blanch–air cool	14.0	9.0	18.6

Differences in both steam versus water blanching and air versus water cooling are significant at the 5% level. Adapted from Cumming *et al.* (1981).

- the surface-area-to-volume ratio of the pieces of food
- method of blanching
- time and temperature of blanching (lower vitamin losses at higher temperatures for shorter times)
- the method of cooling
- the ratio of water to food (in both water blanching and cooling).

Losses of ascorbic acid are used as an indicator of food quality, and therefore the severity of blanching (Table 10.2).

10.3.2 Colour and flavour

Blanching brightens the colour of some foods by removing air and dust on the surface and thus altering the wavelength of reflected light. The time and temperature of blanching also influence the change in food pigments according to their *D* value (Chapter 1). Sodium carbonate (0.125% w/w) or calcium oxide are often added to blancher water to protect chlorophyll and to retain the colour of green vegetables, although the increase in pH may increase losses of ascorbic acid. Enzymic browning of cut apples and potatoes is prevented by holding the food in dilute (2% w/w) brine prior to blanching. When correctly blanched, most foods have no significant changes to flavour or aroma, but under-blanching can lead to the development of off-flavours during storage of dried or frozen foods (Chapters 15 and 21). Changes in colour and flavour are described in more detail by Selman (1987).

10.3.3 Texture

One of the purposes of blanching is to soften the texture of vegetables to facilitate filling into containers prior to canning. However, when used for freezing or drying, the time–temperature conditions needed to achieve enzyme inactivation cause an excessive loss of texture in some types of food (for example certain varieties of potato) and in large pieces of food. Calcium chloride (1–2%) is therefore added to blancher water to form insoluble calcium pectate complexes and thus to maintain firmness in the tissues.

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10.5 References

BOMBEN, J. C., DIETRICH, W. C., FARKAS, D. F., HUDSON, J. S. and DE MARCHENA, E. S. (1973) Pilot plant evaluation of individual quick blanching for vegetables. *J. Food Sci.* **38**, 590–594.

BOMBEN, J. C., DIETRICH, W. C., HUDSON, J. S., HAMILTON, H. K. and FARKAS, D.F. (1975) Yields and solids loss in steam blanching, cooling and freezing vegetables. *J. Food Sci.* **40**, 660–664.

CUMMING, D. B., STARK, R. and SANFORD, K. A. (1981) The effect of an individual quick blanching method on ascorbic acid retention in selected vegetables. *J. Food Process Preserv.* **5**, 31–37.

CUMMING, D. B., STARK, R., TIMBERS, G. E. and COWMEADOW, R. (1984) A new blanching system for the food industry, II, Commercial design and testing. *J. Food Process Preserv.* **8**, 137–150.

GILBERT, H., BAXERRES, J. L. and KIM, H. (1980) In: P. Linko, Y. Malkki, J. Olkku and J. Larinkan (eds) *Food Process Engineering*, Vol. 1. Applied Science, London, pp. 75–85.

HALLSTROM, B., SKJOLDERBRAND, C. and TRAGARDH, C. (1988) *Heat Transfer and Food Products*. Elsevier Applied Science, London, pp. 158–242.

LAZAR, M. E., LUND, D. B. and DIETRICH, W. C. (1971) IQB – a new concept in blanching. *Food Technol.* **25**, 684–686.

PHILIPPON, J. (1984) Methods de blanchiment-refroidissement des legumes destines a la congelation. *Sci. Aliments* **4**, 523–550.

SCOTT, E. P., CARROAD, P. A., RUMSEY, T. R., HORN, J., BUHLERT, J. and ROSE, W. W. (1981) Energy consumption in steam blanchers. *J. Food Process Engng* **5**, 77–88.

SELMAN, J. D. (1987) The blanching process, In: S. Thorne (ed.) *Developments in Food Preservation*, Vol. 4. Elsevier Applied Science, Barking, Essex, pp. 205–249.

TIMBERS, G. E., STARK, R. and CUMMING, D. B. (1984) A new blanching system for the food industry, I, Design, construction and testing of a pilot plant prototype. *J. Food Process Preserv.* **2**, 115–133.

WENDT, F. L., LAUBACHER, E. G., WHERRY, R., MARTIN, P. and ROBE, K. (1983) Blancher with heat recovery processes 6 lb of product per lb of steam. *Food Process USA* **44**, 62–63.